Mini-Project 1 AM215: Tennis Match Simulation Summary

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Executive Summary

We extended a continuous-time Poisson process model from individual tennis games to full matches and validated against 2,433 ATP matches from 2024. The two-parameter model (λ_s, λ_r) captures key match dynamics accurately, with overall errors within 7% for most metrics. Tiebreak frequency is underestimated due to uniform-parameter assumptions.

Model Overview

Game Level: Points follow independent Poisson processes, rates λ_s (server) and λ_r (receiver). States include points, deuce, advantage, and absorbing wins. Closed-form expressions exist for all state probabilities.

Match Level: Hierarchical extension simulates games $\rightarrow$ sets $\rightarrow$ matches, including 7-point tiebreaks and best-of-3 sets. Validation compares simulated vs. real ATP matches.

Key Results

Metric	ATP Data	Simulation	Error
Match Duration	107.8 min	115.8 min	7.4%
Total Games	23.2	21.7	6.5%
2-Set Matches	64.6%	65.1%	0.5%
3-Set Matches	34.9%	34.9%	0.0%
Tiebreak Frequency	38.6%	19.2%	50%

Optimal parameters:

$$\lambda_s = 0.0110 \text{ pts/sec}, \quad \lambda_r = 0.0090 \text{ pts/sec}, \quad \text{server advantage } \approx 55\%.$$

Insights and Limitations

- Simple two-parameter model captures a large percentage of match dynamics.
- Duration, total games, and set distribution closely match real data.
- Tiebreak frequency underestimated due to lack of player skill heterogeneity.
- Model ignores momentum, fatigue, surface effects, and first/second serve differences.
- Time scale matters as real match durations include changeovers and delays.

Future Improvements

1. Incorporate player skill heterogeneity to improve tiebreak prediction.
2. Add surface-specific parameters (hard/clay/grass).
3. Model first vs. second serve probabilities.

Dataset

Source: Jeff Sackmann, tennis_atp. **Coverage:** 2,433 matches from 2024

GitHub Link: https://github.com/cmaloney111/am215-milestones/tree/milestone$_2$